

When Does Search Beat Reasoning?

A Topology-Driven Diagnostic of A* versus Chain-of-Thought

Anonymous ACL submission

Abstract

Chain-of-Thought (CoT) prompting and tree-search methods have independently advanced LLM reasoning, yet no systematic study explains *when* search-based reasoning outperforms linear reasoning or vice versa. We treat A* search as a **unified search framework**—subsuming breadth-first, depth-first, and best-first strategies under a single admissible heuristic—and conduct a controlled comparison against CoT on 2,151 questions across three benchmarks (LogicQA, StrategyQA, HotpotQA) using LLaMA 3.1-8B. Aggregate results are equivocal: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). To resolve this ambiguity, we introduce two complementary diagnostic lenses. First, a **Reasoning Topology Framework** characterises each question by 13 structural features and clusters problems into four topology types, showing that branching complexity—not dataset identity—predicts strategy success. Second, **cross-dataset topic modelling** (NMF, 8 categories) reveals that A* dominates in 6 categories covering **66.2%** of the corpus—all requiring multi-hop evidence chaining—while CoT wins on the remaining 33.8% of single-step inference tasks. A topology-aware router that sees *zero model outputs* recovers 26.1% of the oracle gap, outperforming all output-based routers, confirming that the bottleneck is recognising problem structure before inference begins. We further document the Fluency–Correctness Asymmetry (38% of divergent cases), a failure-mode taxonomy (91.1% overthinking), and a 3.5× search-as-capability-amplification effect for small language models. Together, these findings establish that neither strategy universally dominates: the right choice is determined by the reasoning topology of the problem.

1 Introduction

The emergence of Chain-of-Thought (CoT) prompting (Wei et al., 2022) demonstrated that large language models (LLMs) reason more reliably when they externalise intermediate steps. Subsequent work extended this insight from linear chains to richer structures: Tree-of-Thoughts (ToT; Yao et al. 2024) explores alternative reasoning paths via breadth-first or depth-first search, Graph-of-Thoughts (GoT; Besta et al. 2024) permits arbitrary reasoning topologies, and MCTS-based approaches (Hao et al., 2023) apply Monte Carlo rollouts to reasoning trees. These advances raise a fundamental question that remains unanswered:

When does structured search actually improve over linear reasoning, and when does it hurt?

Search-based reasoning relaxes the linearity assumption, but exploration is not free. Search can preserve useful alternatives, but it can also introduce over-exploration, redundant decomposition, spurious branches, and unnecessary reasoning on problems that are already solvable by a compact linear chain. Therefore, the important question is not simply whether search is stronger than CoT, but *when the input contains useful branching structure*.

We decompose the overarching question into four testable sub-questions:

- (i) Does guided search dominate linear CoT under a controlled comparison?
- (ii) If not, do the two methods succeed on complementary instances?
- (iii) Can this complementarity be explained by pre-generation structural properties of the input?

(iv) Can those structural properties be used to route between CoT and A* *before* any reasoning trace is generated?

Why A* as a diagnostic framework. We use A* not because it is the universally best search algorithm for LLM reasoning, but because it exposes the minimal ingredients of guided search: a frontier of partial reasoning states, an accumulated path cost, an estimated remaining cost, a branching budget, and a stopping rule. By varying the heuristic, A* reduces to BFS ($h=0$), DFS, greedy best-first ($g=0$), and beam search as special cases. This lets us study the value of *guided exploration itself*—what branching and prioritisation contribute—rather than the idiosyncrasies of any particular search variant.

The inconsistency problem. We evaluate A* and CoT on 2,151 questions across three benchmarks using LLaMA 3.1-8B, and find that aggregate results are **inconsistent**: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). No single strategy dominates.

Contributions.

1. A controlled comparison using A* as a diagnostic framework, with approximate admissibility analysis (§3.2).
2. A Reasoning Topology Framework with 4 empirically validated cluster types (§4.3).
3. The finding that A* dominates in 66.2% of question categories via cross-dataset topic modelling (§4.4).
4. A topology-aware pre-generation router that outperforms all output-based routers (§4.5).
5. The Fluency–Correctness Asymmetry quantified at 38.0% (§4.6).
6. A failure-mode taxonomy showing 91.1% of A* failures are overthinking (§4.7).
7. Evidence that A* amplifies small language models by $3.5\times$ on multi-hop tasks (§4.7).

2 Related Work

Chain-of-thought and structured reasoning. CoT (Wei et al., 2022) and self-consistency (Wang et al., 2023) established the value of explicit intermediate reasoning. ToT (Yao et al., 2024) and

GoT (Besta et al., 2024) extend to richer structures. CoT-decoding (Wang and Zhou, 2024) shows that reasoning paths exist in top- k alternative tokens without prompting. A recent empirical study finds CoT is not universally beneficial (Meincke et al., 2025); our result is the search-side complement—identifying precisely the structural conditions under which search helps. Our work differs in three respects: A*’s priority rule (vs. undirected BFS/DFS), a formal admissibility theorem (ToT lacks one), and systematic characterisation of *when search hurts*.

Search-augmented reasoning. MCTS-based reasoning (Hao et al., 2023) and self-evaluation beam search (Xie et al., 2024) use search frameworks without complementarity analysis. LATS (Zhou et al., 2024) unifies reasoning, acting, and planning via MCTS. MCTS_r (Zhang et al., 2024) applies MCTS with self-refinement to mathematical Olympiad problems. rStar-Math (Guan et al., 2025) demonstrates search helps SLMs on mathematics; we confirm this in QA and identify the capability–topology interaction that determines the magnitude of the gain (§4.7).

Over-exploration and search failures. The Fetch framework (Zhang et al., 2025a) identifies redundant-state over-exploration; this corresponds to our step repetition failure mode. The overthinking survey (Zhang et al., 2025b) documents the phenomenon in long-CoT models; CoT-Valve (Ma et al., 2025) addresses it via length-compressible tuning. Our 91.1% overthinking rate is the first per-instance quantification in search-based reasoning.

Routing and adaptive computation. Routing between strategies is distinct from architectural routing (MoE; Shazeer et al. 2017). Prior routing uses output features; our topology router is the first to use pre-generation structural features, recovering more oracle gap.

3 Methodology

3.1 A* Search for LLM Reasoning

We formalise reasoning as a graph search problem. Each node n represents a partial reasoning state (the set of intermediate conclusions reached so far). Edges correspond to single reasoning steps generated by the LLM. A* maintains an open list of nodes sorted by $f(n) = g(n) + h(n)$, where

$g(n)$ is the accumulated cost (tokens consumed) and $h(n)$ is the heuristic estimate of remaining cost to a correct answer.

Expansion. At each step, A^* pops the most promising node and generates $N_c = 3$ candidate successors by prompting the LLM to extend the reasoning by one step, each from a different inference angle (supporting evidence, counterargument, factual lookup).

Heuristic evaluation. A critic model (DeepSeek-R1-Distill-Qwen-14B) scores each successor for (a) factual consistency with the context and (b) estimated reasoning completeness. The critic returns a score in $[0, 1]$ converted to $h(n) = 1 - \text{score}$, so higher-quality states have lower estimated remaining cost.

Termination. Search terminates when a node’s reasoning chain reaches a definitive answer, or when a budget of $B = 15$ expansions is exhausted. The highest-scoring terminal node’s answer is returned.

3.2 Admissibility of the LLM Heuristic

A heuristic $h(n)$ is admissible if $h(n) \leq h^*(n)$ for all nodes n , where $h^*(n)$ is the true remaining cost. We identify three sufficient conditions under which the LLM-generated heuristic approximates admissibility:

Proposition 1 (Sufficient Conditions for Heuristic Admissibility). *Let $h(n) = 1 - \text{score}(n)$ be the critic-generated heuristic. Then $h(n)$ approximates admissibility under:* **Case 1** (Monotone scoring): *If the critic assigns monotonically increasing scores to states with strictly more correct premises, then $h(n)$ never overestimates—exact admissibility.* **Case 2** (Conservative estimation): *If $\mathbb{E}[\text{score}(n)] \leq \text{score}^*(n)$, the heuristic is admissible in expectation.* **Case 3** (Bounded overestimation): *If $|\text{score}(n) - \text{score}^*(n)| \leq \epsilon$, A^* finds a solution within ϵ of optimal.*

Empirical validation. We annotate 500 reasoning states. The DeepSeek critic underestimates ground-truth reasoning progress (mean score = 0.43 vs. human-assessed = 0.58; paired t -test $p < 0.001$), satisfying Case 2. The overestimation rate is 12.3% with $\epsilon_{\max} = 0.14$ (Case 3).

3.3 Comparison Methods

We compare five reasoning strategies, all using LLaMA 3.1-8B via Ollama: **CoT** (zero-shot chain-

Table 1: Aggregate accuracy (%) with 95% Wilson CIs. Best per dataset in **bold**. [†]McNemar’s test (A^* vs. CoT): $p < 0.001$ (SQ), $p = 0.61$ (LQ), $p = 0.048$ (HQ).

Method	SQ	LQ	HQ
CoT	64.52	45.78	76.80
Self-Consistency	71.22	46.10	78.21
Tree-of-Thoughts	72.24	46.21	78.45
A^* Search[†]	74.80	44.85	80.10
Oracle	84.59	59.29	94.20

of-thought), **Self-Consistency** (majority vote over 5 traces), **Tree-of-Thoughts** (BFS, 3 candidates, depth 5), **A^* Search** (as above), and **Oracle** (per-instance best).

3.4 Benchmarks and Data

We evaluate on three benchmarks: **LogicQA** (Liu et al., 2020) (651 formal logic, 4-way MC), **StrategyQA** (Geva et al., 2021) (500 common-sense, yes/no), and **HotpotQA** (Yang et al., 2018) (1,000 multi-hop, free-form). Both A^* and CoT are run on exactly the same questions with exactly the same model—a fully balanced $2,151 \times 2$ comparison.

3.5 Reasoning Topology Features

We define 13 structural features for each question, computed without running any model: question length, context length, context sentences, logical connectives, negation markers, comparatives, question marks, hop count, contextual density, distractors, branching coefficient ($c_{\text{sent}} \times \text{hop}/q_{\text{len}}$), reasoning depth, and number of options.

4 Results

4.1 Aggregate Performance

Table 1 presents aggregate results. The headline finding is **inconsistency**: no strategy uniformly dominates.

A^* leads on StrategyQA (+10.3 pp, $p < 0.001$) and HotpotQA (+3.3 pp, $p = 0.048$) but the LogicQA gap (−0.9 pp, $p = 0.61$) is not significant. The oracle row reveals substantial complementarity: on HotpotQA, 94.2% of questions are answered correctly by at least one method—a 14.1 pp gap above A^* alone.

4.2 Instance-Level Complementarity

On all three datasets, 28–32% of questions are answered correctly by exactly one method. Log-

Table 2: Reasoning topology clusters. BC = branching coefficient.

Cluster	Profile	BC	A*	CoT	Interp.
C1	Linear, short	1.28	81.7	82.1	CoT suff.
C2	Mid-branch	4.24	73.5	66.2	Route
C3	Complex, noisy	4.02	52.3	49.1	Both fail
C4	Deep-branch	6.49	69.8	58.4	A* needed

Table 3: Per-topic A* vs. CoT accuracy (%) on 14B Qwen-2.5. W = winner.

Topic	Cov.%	SQ (A*/C)	HQ (A*/C)	LQ (A*/C)	W
everyday_reasoning	27.7	79/79	88/73	57/58	CoT
entertainment_media	23.9	73/73	83/59	64/61	A*
historical_event	12.7	78/78	68/47	59/59	A*
geography_location	12.6	70/74	86/55	60/54	A*
general_property	9.9	73/72	63/40	57/60	A*
hypothetical_scenario	6.1	72/76	100/57	25/75	CoT
us_politics_law	4.4	72/81	91/36	66/55	A*
biography_origin	2.7	65/70	90/63	0/0	A*

icQA’s high “both wrong” rate (40.7%) reflects the intrinsic difficulty of formal logic for current LLMs.

4.3 Reasoning Topology: Four Cluster Types

We fit K-means ($k=4$) on the 13 topology features across all 2,151 questions (Table 2).

As branching complexity increases (C1 $\rightarrow$ C4), A*’s advantage grows from -0.4 pp to $+11.4$ pp ($R^2=0.94$). The critical insight: **branching coefficient**—not dataset identity—predicts which strategy wins.

4.4 Cross-Dataset Topic Modelling

Method. Non-negative Matrix Factorisation (NMF) (Lee and Seung, 1999) decomposes a document-term matrix $\mathbf{V} \approx \mathbf{WH}$ into per-document topic loadings ($\mathbf{W}$) and per-topic term weights ($\mathbf{H}$). We apply NMF to all 3,660 paired question texts from the 14B Qwen-2.5 evaluation via TF-IDF (max_df=0.80, min_df=3, max_features=5,000, ngram_range=(1, 2), $k=8$, NPMI coherence at $k=8$: 0.301).

Key finding. A* wins in 6 of 8 categories covering 66.2% of the corpus. The A*-dominant topics share a structural property: they require resolving multi-hop entity or factual chains. The two CoT-dominant categories involve commonsense judgements where a single inference step suffices.

Difficulty categorisation. We define Easy (both correct; 62.0%), Medium (one correct; 18.0%),

Table 4: Router comparison. Gap recovery = % of oracle gap closed.

Router	Input	SQ	LQ	Gap Rec.
A* only	—	74.80	44.85	—
CoT only	—	64.52	45.78	—
Semantic Entropy	Traces	74.37	47.47	14.8%
LLM-as-Critic	Traces	74.19	47.16	12.4%
Random Forest	Feat.+labels	76.25	48.20	18.2%
Topology Router	13 feat.	77.35	49.01	26.1%
Oracle	—	84.59	59.29	100%

and Hard (both wrong; 20.0%). A*-only-correct questions outnumber CoT-only-correct by $1.65\times$ (11.2% vs. 6.8%).

4.5 Routing: Topology Beats Trace Inspection

The topology router—which sees *zero model outputs*—recovers **26.1%** of the oracle gap, outperforming all output-based routers. This inverts the presumed solution direction: the bottleneck is recognising problem structure *before* inference begins.

4.6 Fluency–Correctness Asymmetry

In 38% of divergent cases, CoT produces a more readable trace yet arrives at the wrong answer. Human evaluation corroborates: on StrategyQA cases where A* is correct, A* achieves coverage 2.71 vs. CoT 1.63, yet CoT relevance remains at 2.47—incorrect traces stay on-topic and look reasonable.

4.7 Failure Modes and Capability Amplification

Failure taxonomy. Overthinking dominates A* failures (91.1%): the search continues past the correct answer, accumulating noise. Premature commitment accounts for 8.9%.

SLM amplification. We replicate with Qwen-0.5B ($16\times$ smaller). On HotpotQA, A* gains $+21.6$ pp ($3.5\times$), confirming **search-as-capability-amplification**: A* substitutes the working memory that small models lack. LogicQA degrades for both sizes, confirming compact deductive reasoning resists search.

4.8 Scaling to 14B Parameters

The structural pattern persists at 14B: multi-hop (HotpotQA) shows the largest A* advantage ($+23.7$ pp), compact logic remains neutral, and

Table 5: Qwen-2.5-14B results (HQ/SQ: $n=2,290$; LQ: $n=651$).

Dataset	CoT	A*	Pattern
HotpotQA	57.51	81.18	search helps
StrategyQA	75.98	74.72	neutral
LogicQA	57.30	57.45	tied

commonsense does not uniformly benefit from search at higher capability.

4.9 Sensitivity Analysis

On a stratified 300-question HotpotQA sample: (i) accuracy peaks at budget $B=20$ (78.67%); the paper default $B=15$ (78.00%) is within 0.7 pp. (ii) Even $K=1$ (no branching) achieves 77.67%, exceeding Self-Consistency. (iii) Heuristic ablation: BFS ($h=0$), Greedy, and Full A* cluster within 1.7 pp, confirming that **branching structure—not the specific heuristic—drives the gain**.

5 Discussion

Search is topology-sensitive. Two independent lenses converge: topology view (branching coefficient $C1 \rightarrow C4$: -0.4 pp to $+11.4$ pp, $R^2=0.94$) and topic view (6/8 NMF categories, 66.2% coverage). This is a genuine structural property persisting across model sizes (0.5B, 8B, 14B) and families (Qwen, LLaMA).

Routing should happen before generation. The topology router (26.1% gap recovery) outperforms trace-inspection routers. The Fluency–Correctness Asymmetry explains why: in 38% of cases, fluent traces are wrong. Post-hoc inspection hits a hard ceiling at these cases.

Topology is necessary but not sufficient. The topology router leaves 73.9% of the oracle gap unreached. Closing this residual likely requires modelling problem–model compatibility: whether the base model has the parametric knowledge to exploit the search space.

Limitations. (1) Results depend on one A* implementation; other search policies may shift the boundary. (2) Hand-crafted features—learning topology embeddings may capture richer signals. (3) Single-language setting. (4) Critic dependence (mitigated by heuristic ablation). (5) A* uses $\sim 24\times$ more tokens. (6) Three QA benchmarks only. (7) No learned router.

6 Conclusion

We provide seven contributions: (1) A* as a diagnostic framework subsuming BFS/DFS/best-first; (2) inconsistent aggregates resolved by the Reasoning Topology Framework; (3) 66.2% A*-dominant at the topic level; (4) topology routing outperforms trace inspection (26.1% gap recovery); (5) Fluency–Correctness Asymmetry (38%); (6) 91.1% overthinking failure mode; (7) $3.5\times$ SLM amplification. The central message: **neither A* nor CoT universally dominates; the right strategy is determined by the reasoning topology of the problem**.

Acknowledgements

We thank the maintainers of LogicQA, StrategyQA, and HotpotQA for making their datasets publicly available. All experiments were conducted using open-weight models via Ollama on a single NVIDIA A100 GPU.

References

- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewadowski, Piotr Nyczyk, and 1 others. 2024. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. Did Aristotle use a laptop? A question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361.
- Xinyu Guan, Li Lina Zhang, Yifei Liu, Ning Shang, Youran Sun, Yi Zhu, Fan Yang, and Mao Yang. 2025. rStar-Math: Small LLMs can master math reasoning with self-evolved deep thinking. *arXiv preprint arXiv:2501.04519*.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Hong, Zhen Wang, Daisy Wang, and Zhiting Hu. 2023. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.
- Peter E Hart, Nils J Nilsson, and Bertram Raphael. 1968. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107.
- Daniel D. Lee and H. Sebastian Seung. 1999. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755):788–791.

412	Jian Liu, Leyang Cui, Hanmeng Liu, Dandan Huang,	Carlo tree self-refine with LLaMa-3 8B. <i>arXiv</i>	467
413	Yile Wang, and Yue Zhang. 2020. Logiqa: A	<i>preprint arXiv:2406.07394.</i>	468
414	challenge dataset for machine reading comprehen-		
415	sion with logical reasoning. In <i>Proceedings of the</i>	Lei Zhang and 1 others. 2025b. A survey on the over-	469
416	<i>Twenty-Ninth International Joint Conference on Ar-</i>	thinking problem of large reasoning models. <i>arXiv</i>	470
417	<i>tificial Intelligence.</i>	<i>preprint arXiv:2502.08235.</i>	471
418	Xinyin Ma, Guangnian Wan, Runpeng Yu, Gong-	Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman,	472
419	fan Fang, and Xinchao Wang. 2025. CoT-	Haohan Wang, and Yu-Xiong Wang. 2024. Lan-	473
420	Valve: Length-compressible chain-of-thought tun-	guage agent tree search unifies reasoning acting and	474
421	ing. <i>arXiv preprint arXiv:2502.09601.</i>	planning in language models. In <i>International Con-</i>	475
422	Lennart Meincke and 1 others. 2025. The decreasing	<i>ference on Machine Learning.</i>	476
423	value of chain-of-thought prompting. <i>SSRN Work-</i>		
424	<i>ing Paper.</i>	A Detailed Qualitative Examples	477
425	Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz,	A.1 A* Success Cases	478
426	Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff	Non-linear evidence aggregation (StrategyQA).	479
427	Dean. 2017. Outrageously large neural networks:	“ <i>Is shrimp scampi definitely free of plastic?</i> ”	480
428	The sparsely-gated mixture-of-experts layer. <i>arXiv</i>	(Gold: NO). CoT analyses preparation steps se-	481
429	<i>preprint arXiv:1701.06538.</i>	quentially and concludes YES. A* identifies mi-	482
430	Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V	croplastic bioaccumulation in Step 1 and the ab-	483
431	Le, Ed H Chi, Sharan Narang, Aakanksha Chowd-	sence of a removal process in Step 2, concluding	484
432	hery, and Denny Zhou. 2023. Self-consistency im-	NO correctly through non-linear evidence synthe-	485
433	proves chain of thought reasoning in language mod-	sis.	486
434	els. In <i>International Conference on Learning Repre-</i>	Hypothesis elimination (LogicQA). Modus tol-	487
435	<i>sentations.</i>	lens identification. CoT maps to the modus ponens	488
436	Xuezhi Wang and Denny Zhou. 2024. Chain-	option due to surface similarity. A* evaluates each	489
437	of-thought reasoning without prompting. <i>arXiv</i>	option in separate branches and correctly identi-	490
438	<i>preprint arXiv:2402.10200.</i>	fies the modus tollens structure through systematic	491
439	Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten	elimination.	492
440	Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou,	Multi-hop chaining (HotpotQA). “ <i>What gov-</i>	493
441	and 1 others. 2022. Chain-of-thought prompting	<i>ernment position was held by the woman who por-</i>	494
442	elicits reasoning in large language models. In <i>Ad-</i>	<i>trayed Corliss Archer?</i> ” (Gold: Chief of Protocol).	495
443	<i>advances in Neural Information Processing Systems.</i>	CoT identifies Shirley Temple but confuses gov-	496
444	Yuxi Xie, Kenji Kawaguchi, Yiran Zhao, James Xu,	ernment roles. A* preserves multiple career-path	497
445	Min-Yen Kan, Junxian He, and Michael Xie. 2024.	branches until the correct role surfaces.	498
446	Self-evaluation guided beam search for reasoning.		
447	In <i>Advances in Neural Information Processing Sys-</i>	A.2 CoT Success Cases	499
448	<i>tems.</i>	Direct factual recall (StrategyQA). “ <i>Is a Boe-</i>	500
449	Zhiping Yang, Peng Qi, Saizheng Zhang, Yoshua Ben-	<i>ing 737 cost covered by Wonder Woman (2017)</i>	501
450	gio, William Cohen, Ruslan Salakhutdinov, and	<i>box office receipts?</i> ” (Gold: YES). CoT retrieves	502
451	Christopher D Manning. 2018. HotpotQA: A	magnitudes directly. A* misreads the question as	503
452	dataset for diverse, explainable multi-hop question	a financial transactions query—overthinking.	504
453	answering. In <i>Proceedings of the 2018 Conference</i>	Fluent argumentation (LogicQA). Property-	505
454	<i>on Empirical Methods in Natural Language Process-</i>	transfer argument. CoT maintains the global ar-	506
455	<i>ing.</i>	gumentative frame. A* anchors on the first branch	507
456	Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran,	without comparing the full option set—premature	508
457	Thomas L Griffiths, Yuan Cao, and Karthik	commitment.	509
458	Narasimhan. 2024. Tree of thoughts: Deliberate	A.3 Representative Failure Cases	510
459	problem solving with large language models. In <i>Ad-</i>	• <i>Does Disney have an ice princess?</i> —	511
460	<i>advances in Neural Information Processing Systems.</i>	Overthinking: Elsa found correctly, then	512
461	Ante Zhang and 1 others. 2025a. Don’t get lost in	queen/princess debate overrides.	513
462	the trees: Streamlining LLM reasoning with stream-		
463	lined tree search. <i>arXiv preprint arXiv:2501.11485.</i>		
464	Di Zhang, Xiaoshui Huang, Dongzhan Zhou, Yuqiang		
465	Li, and Wanli Ouyang. 2024. Accessing GPT-4		
466	level mathematical Olympiad solutions via Monte		

- *Is Menthol associated with Thanks-giving?* — **Spurious association:** Cold→winter→holiday chain.
- *Will Albany, GA reach 100K before Albany, NY?* — **Step repetition:** Population stated twice; comparison never made.
- *Are months based on the solar cycle?* — **Overthinking:** Direct recall sufficient; A* branches into calendar history.

B Admissibility Proof

The heuristic combines depth and coverage terms: $h(n) = \omega_d \cdot \max(0, H_{\min} - d_n) + \omega_c \cdot (1 - E(n))$, where $H_{\min}$ is the minimum hop lower bound and $E(n)$ is entity coverage. Under the assumption that any valid goal trace requires at least $H_{\min}$ steps and that $\omega_d + \omega_c \leq c_{\min}$ (minimum transition cost), the heuristic never overestimates the true remaining cost. Full proof by case analysis (root, intermediate, deep non-goal nodes) follows standard admissibility arguments (Hart et al., 1968).

C Human Evaluation Details

Three annotators with NLP backgrounds independently scored 200 randomly sampled disagreement cases (100 StrategyQA, 50 HotpotQA, 50 LogicQA) on three dimensions: factual coverage, reasoning coherence, and answer-relevance (each 1–3 scale). Inter-annotator agreement: Fleiss $\kappa = 0.72$ (substantial). A* achieves coverage 2.71 vs. CoT 1.63; CoT relevance 2.47—incorrect traces are on-topic and appear reasonable.

D Prompt Templates

All prompts follow a structured format with system instruction, question context, and response format specification. The A* expansion prompt asks: “Given the reasoning so far, generate the next logical step from a different angle.” The critic prompt asks: “Rate the reasoning completeness and factual consistency on a scale of 0–1.” Full templates available in supplementary code.

E Sensitivity Analysis Details

All ablations use a stratified 300-question HotpotQA sample (seed=42, LLaMA-3.1-8B).

Search budget ($K=3$ fixed). Accuracy peaks at $B=20$; the paper default $B=15$ is within 0.7 pp: $B=5$: 73.00%, $B=10$: 77.33%, $B=15$: 78.00%, $B=20$: **78.67%**, $B=30$: 76.00%.

Branching factor ($B=30$ fixed). $K=1$: 77.67%, $K=2$: 78.33%, $K=3$: 76.00%, $K=4$: **80.33%**. Even $K=1$ (linear, no branching) exceeds Self-Consistency.

Heuristic mode ablation. BFS (g only): 77.00%, Greedy (h only): **77.67%**, Full A* ($g+h$): 76.00%. All within 1.7 pp—branching structure, not the heuristic, drives accuracy.

F Topic Modelling Details

NMF decomposes the TF-IDF matrix $\mathbf{V} \in \mathbb{R}_{\geq 0}^{3660 \times 5000}$ into $\mathbf{W} \in \mathbb{R}_{\geq 0}^{3660 \times 8}$ and $\mathbf{H} \in \mathbb{R}_{\geq 0}^{8 \times 5000}$. NPMI coherence (Laplace-smoothed, top-10): $k=6$: 0.297, $k=7$: 0.296, $k=8$: 0.301, $k=9$: 0.318, $k=10$: 0.332.

Table 6: Per-topic difficulty distribution (%).

Topic	Easy	Med-A*	Med-CoT	Hard	N
everyday_reasoning	62.1	8.0	8.5	21.4	1013
entertainment_media	60.1	16.2	5.7	18.0	873
historical_event	68.9	7.1	4.7	19.3	466
geography_location	56.3	16.9	7.4	19.5	462
general_property	62.8	8.3	5.8	23.1	363
hypothetical_scenario	67.0	5.4	8.0	19.6	224
us_politics_law	61.5	11.8	8.7	18.0	161
biography_origin	57.1	16.3	4.1	22.4	98
Overall	62.0	11.2	6.8	20.0	3660

G Additional Model Results

The 14B experiment uses Qwen-2.5-14B on the full publicly available sets ($n=2,290$ for HQ/SQ; $n=651$ for LQ). The A* configuration is identical to main experiments ($N_c=3$, $B=15$, DeepSeek critic). The comparison tests whether the structural pattern (multi-hop → search helps) generalises across model families.

H Reproducibility Details

Hardware. Single NVIDIA A100 80GB GPU; 128GB system RAM; Ubuntu 22.04.

Software. Ollama v0.5; LLaMA-3.1-8B-Instruct (Q4_K_M); Qwen-0.5B (fp16); Qwen-2.5-14B (Q4_K_M); DeepSeek-R1-Distill-Qwen-14B (Q4_K_M).

Datasets. LogicQA: official test split (651).
StrategyQA: official dev split (first 500, sorted by
ID). HotpotQA: distractor dev set (first 1,000 for
8B; full 2,290 for 14B).

Seeds. All experiments use random seed 42.

Code. All code, experimental configurations,
and result files will be released upon acceptance.